FLUTTER APPLICATION DEVELOPMENT PROGRAMME

Project Specification and Completion Guide

Group Project

Item	Details
Programme	Flutter Application Development, 23 sessions, 31 August to 21 October 2026
Delivered by	IITM Pravartak and BodhBridge Flutter Lab
Prepared for	Students of KHIT Guntur
Document purpose	Specification of the group project, to be completed after the training sessions
Version	1.0, October 2026

1. Purpose of this document

The training sessions are complete. This document sets out the group project that each team must now finish: what the application must contain and what must be submitted.

The document describes requirements only. It contains no solutions or code. Build the project using what was taught in the sessions and the session handbooks already shared.

2. Overview

Item	Details
What	An Android application on one approved topic
Who	Teams of two or three students
Builds on sessions	11 to 23: state, forms, navigation, themes, APIs, Provider, saving on the device, Git, testing and release
Delivered as	GitHub repository, release APK and a live demonstration

3. Technical requirements

Area	Requirement
Framework	Flutter, stable channel, version 3.35 or newer. Language Dart with null safety.
Platform	Android. The application must install and run on a physical Android phone from a release APK.
Tools	Android Studio with the Flutter and Dart plugins, JDK 17, an Android emulator for development.
State management	Provider, using a class that extends ChangeNotifier for data shared between screens.
Project structure	Code organised inside lib into models, services, screens and widgets folders, with one main class per file.
Version control	A GitHub repository, with regular commits from the start of work, not a single upload at the end.
Code quality	No hard coded colours outside the theme. No unused files or commented out blocks of old code. The project must run without errors from a fresh clone.
Originality	Each student must be able to explain any part of their own code.

4. Group project

4.1 Team rules

- Teams of two or three students.
- Each team chooses one topic from section 5.3. Teams that choose the same topic must build clearly different applications.
- Every member owns at least one screen or one major part, such as the data service or the form, and this is recorded in the README.
- Every member commits to the shared repository regularly.

4.2 Mandatory requirements, all topics

ID	Requirement	Detail
GP 1	Four screens minimum	At least a list screen, a detail screen, a form screen and one further

ID	Requirement	Detail
		screen of the team’s choice. All reachable by tapping.
GP 2	Real data	Data is either fetched from a public API over the internet, or created by the user and saved on the device so it survives a restart. Hard coded sample lists are not accepted in the final version.
GP 3	Four screen states	The main list screen correctly shows a loading indicator, an error message with a retry option, an empty message, and the data.
GP 4	Validated form	The form screen checks every field before saving and shows a clear message for each invalid entry.
GP 5	Detail view	Tapping an item opens a detail screen showing more information than the list row.
GP 6	Shared state	Application data is held in at least one Provider store and read by more than one screen.
GP 7	Theme	A consistent theme from one seed colour, used across every screen, and the application’s own name and icon.
GP 8	Robustness	No crashes and no unhandled red error screens when tested with empty input, very long text, no internet connection, repeated fast taps and the phone turned sideways.
GP 9	Repository	A GitHub repository with commits from every member from the start of work, and a README as described in the submission section.
GP 10	Release build	A release APK that installs and runs on a physical phone, including internet access where the application uses an API.
GP 11	Test	At least one widget test that runs and passes with the flutter test command.

4.3 Approved topics

Each topic lists the minimum scope. Teams may add features once the mandatory requirements are met, but should not start optional features before then.

Topic	Minimum scope
Recipe application	Browse a list of recipes, open a recipe to see ingredients and steps, add a personal recipe through a form, mark favourites. Data from a public recipe API or saved on the device.
News reader	List of current articles with title, source and date, a detail screen for each article, filtering by category, and a form to save personal notes or bookmarks. Data from a public news API.
Expense tracker	Add expenses with amount, category, date and note through a validated form. List of expenses, a detail screen, and a summary screen with totals per category. Data saved on the device.
Job board	List of job openings with role, company and location, a detail screen with the full description, filtering by location or type, and an application form with validation. Data from a public API or saved on the device.
Weather application	Current weather for a searched city, a detail screen with a multi day forecast, a list of saved cities, and a settings screen for units. Data from a public weather API.
Quiz application	List of quiz categories, a question screen with options and score tracking, a results screen, and a form to add custom questions. Questions from a public quiz API or saved on the device. Best scores kept after restart.
College notice board	List of notices with title, date and department, a detail screen, filtering by department, and a form to post a notice with validation. Data saved on the device or from a public API.

Some public APIs require a free key. If a key is used, it must not be committed to the public repository.

5. Submission requirements

- Link to the team’s public GitHub repository, with commits from every member.
- The release APK file.
- A README containing the application name and topic, a short description, at least three screenshots, the data source used, the steps to run the project, and each member’s name with the parts they built.
- A defect log listing each defect found during testing, the steps to reproduce it, and whether it was fixed.
- A five minute live demonstration on a physical phone, followed by questions.

5.1 Demonstration format

Time	Content
0:00 to 0:30	Who the application is for and what problem it solves.
0:30 to 3:30	The primary flow live on the phone, including a rejected form entry and the error or empty state shown on purpose.
3:30 to 4:30	One technical problem the team faced and how it was solved. Every member speaks during the demonstration.
4:30 to 5:00	The GitHub repository commit history and the APK file.

Slide presentations are not accepted in place of the working application.